

Doping Stability Report -- Mg in NaCoO₂ at 200 C

Generated: 2026-05-30 11:14 Host: NaCoO₂ Dopant: Mg

Executive Summary

Mg preferentially occupies the Co layer. The formation energy difference between the two layers is +3.061 eV, strongly favouring the Co site. MD confirms this preference: on the Co site the dopant MSD plateau slope is 0.00422 Å²/ps (stable), compared to 0.00087 Å²/ps on the Na site. The Na-site E_f is approximate due to a charge surplus of +1 that cannot be modelled in CHGNet.

Site	E _f (eV)	Charge	Compensation	MSD pass	Coord pass	Verdict
Co	-2.561	-1	No	PASS	PASS	STABLE
Na	+0.500	+1	Yes	PASS	FAIL	STRUCTURAL COLLAPSE

Mg at the Co layer -- STABLE

Charge situation

Mg replaces Co with a charge mismatch of -1. To restore charge neutrality, 1 Na atom(s) were removed from the supercell (furthest from the dopant site). The formation energy accounts for the Na-vacancy term.

Formation Energy (thermodynamic stability)

Formation energy E _f	-2.5612 eV	PASS	< 1.0 eV to pass
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E_f = -2.561 eV. This value is strongly negative, indicating that Mg incorporation is thermodynamically very favorable at the Co site.

Molecular Dynamics Stability (200 C, 25 ps)

MSD slope (late-time)	0.00422 Å ² /ps	PASS	< 0.005 to pass (plateau = stable)
Final MSD	0.239 Å ²	----	
Max displacement	0.971 Å	----	
Coordination (min/mean/max)	6 / 6.0 / 7	PASS	relaxed = 6, tolerance +/-1
Mean bond length (MD vs relaxed)	2.031 Å (relaxed: 2.0)	----	
Lattice volume change	+0.00 %	PASS	< +/-5% to pass

Late-time MSD slope = 0.00422 Å²/ps. The slope is small and below the migration threshold -- the dopant is site-stable. Final MSD = 0.239 Å²; maximum displacement from starting position = 0.97 Å.

Coordination fluctuated between 6 and 7 (mean 6.0), within the +/-1 tolerance of the relaxed value (6). This is consistent with normal thermal vibration. Mean nearest-neighbour distance during MD = 2.031 Å.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: STABLE

Mg is thermodynamically favorable and kinetically stable on the Co site at 200 C. All four stability criteria pass: formation energy is favorable, the dopant does not migrate during MD, its coordination environment is

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preserved, and the host lattice volume is unchanged.

? **WARNING:** Charge deficit (-1) cannot be compensated in CHGNet (would require adding Na interstitials or oxidising Co). Running single substitution without compensation. E_f is approximate.

Mg at the Na layer -- STRUCTURAL COLLAPSE

Charge situation

Mg replaces Na with a charge surplus of +1. CHGNet cannot model this compensation (would require Co oxidation or electron addition). The formation energy below is approximate.

Formation Energy (thermodynamic stability)

Formation energy E_f	+0.5000 eV	PASS	< 1.0 eV to pass
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E_f = +0.500 eV (approximate -- charge surplus not modelled in CHGNet). This value is moderately positive, indicating that Mg incorporation is possible but endothermic; may require non-equilibrium synthesis at the Na site.

Molecular Dynamics Stability (200 C, 25 ps)

MSD slope (late-time)	0.00087 Å²/ps	PASS	< 0.005 to pass (plateau = stable)
Final MSD	0.031 Å²	----	
Max displacement	0.598 Å	----	
Coordination (min/mean/max)	4 / 5.9 / 7	FAIL	relaxed = 6, tolerance +/-1
Mean bond length (MD vs relaxed)	2.141 Å (relaxed: 2.1)	----	
Lattice volume change	+0.00 %	PASS	< +/-5% to pass

Late-time MSD slope = 0.00087 Å²/ps. The slope is near zero, confirming the dopant is tightly confined to its site. Final MSD = 0.031 Å²; maximum displacement from starting position = 0.60 Å.

Coordination fluctuated between 4 and 7 (mean 5.9), deviating by more than +/-1 from the relaxed value (6). This may indicate structural distortion, or that the bond-length cutoff is too close to the actual bond length -- check the coordination_cutoff_A setting. Mean nearest-neighbour distance during MD = 2.141 Å.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: STRUCTURAL COLLAPSE

The coordination criterion failed while MSD and volume both pass, suggesting a possible cutoff artifact rather than a true structural collapse. The Mg atom barely moves (MSD = 0.031 Å²) but the coordination count fluctuates because the bond length at the Na site is close to the analysis cutoff. Consider increasing coordination_cutoff_A and re-running analysis.

Site Preference Comparison

The Co site is preferred by 3.061 eV over the Na site. Formation energy: -2.561 eV (Co) vs +0.500 eV (Na). Mg preferentially occupies the Co layer. The formation energy difference between the two layers is +3.061 eV, strongly favouring the Co site. MD confirms this preference: on the Co site the dopant MSD plateau slope is 0.00422 Å²/ps (stable), compared to 0.00087 Å²/ps on the Na site. The Na-site E_f is

approximate due to a charge surplus of +1 that cannot be modelled in CHGNet.

Caveats and Limitations

? CHGNet is charge-neutral.

The potential does not model oxidation states, polaron formation, or electron/hole localisation. Formation energies for aliovalent dopants (charge mismatch $\neq 0$) are approximate even when structural compensation defects are included.

? No Freysoldt/Kumagai correction.

Electrostatic finite-size corrections for charged defects require the host dielectric tensor and the DFT electrostatic potential, neither of which is available from a neural-network potential. For publication-quality E_f of charged defects, follow up with DFT using doped or pymatgen-analysis-defects.

? Metal-rich reference state.

Chemical potentials are computed from elemental ground-state structures. Synthesis under oxide-rich or other conditions shifts all E_f values by additive constants; the relative ordering of sites is unaffected.

? MD timescales are short (25 ps default).

This is sufficient to assess local site stability and rapid migration, but cannot resolve slow diffusion mechanisms. A non-plateauing MSD confirms instability; a plateau does not guarantee stability on longer timescales.

? Energy rankings are more trustworthy than absolute values.

CHGNet is a universal MLIP trained on r2SCAN DFT. Formation-energy errors for transition-metal oxides are typically 50-150 meV/atom. Use relative orderings (which site is preferred?) as the primary result.